SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Error Analysis Task

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 3 – Error Analysis Task
Weightage:	35% of CIA	**Duration / Marks:**	60 mins / 100 marks
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
Student Name:	S Sandhiya	**Reg. No.:**	192321015
Section / Batch:	Slot D	**Date:**	10-10-2026

Code of Conduct

I S Sandhcertify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S Sandhiya

Instructions

- Read each question carefully; answer all four questions (100 in total). Off-Class Assessment Tool.
- Each question gives a student's faulty work. Do not solve from scratch: identify every error (cite it by statement, component, iteration or step number), explain its root cause, correct it with justification and apply the corrected concept. An error merely carried forward counts once.
- Follow the conventions stated in each question (alphabetical expansion, explored set, goal test on removal from the frontier, tie-breaking) and present tables neatly.
- Marks follow the Error Analysis rubric: error identification, root cause analysis, correction and justification, application of concepts, and clarity and presentation.
- Upload in GitHub and GCR (question paper with Reg. No., Name, answers, rubrics and evaluation page at the end).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Four approaches to AI, Turing Test, eras and milestones	**Q1**
2. Intelligent Agents: Agents and Environments	Agent, percepts, actions, agent function and program, PEAS	**Q1**
3. The Concept of Rationality	Performance measure, percept sequence, rational vs omniscient	**Q1**
4. The Nature of Environments	Task-environment properties and their classification	**Q1**
5. The Structure of Agents	Simple reflex and model-based reflex agent programs	**Q1**
6. Problem-Solving Agents	Goal formulation, problem formulation, search, execution	**Q2**
7. Problem Formulation	States, initial state, actions, goal test, path cost; belief states	**Q2, Q4**
8. Searching for Solutions	Search graph, frontier, explored set, node expansion	**Q3**
9. Uninformed Search Strategies	Error tracing of BFS, DFS and uniform-cost search	**Q3**
10. Search with Partial Information	Sensorless problems, belief states, percept update	**Q4**

Annexure A – Error Analysis Task

Question 1: Foundations of AI, Intelligent Agents and Rationality – Error Analysis of Concept Notes

Topic: Foundations and history of AI, agents and environments, rationality, nature of environments, structure of agents

A junior engineer prepared the revision notes below on the foundations of AI and intelligent agents. The notes contain errors; some statements may be correct. Analyse them statement by statement.

No.	Statement in the notes
S1	The Turing Test (1950) judges a machine to be intelligent if its internal reasoning always follows the laws of logic. It is therefore the standard example of the “thinking rationally” approach to AI.
S2	The “AI winters”, periods of reduced funding and interest, occurred in the 1950s, before the 1956 Dartmouth workshop. That workshop then revived the field and coined the term “artificial intelligence”.
S3	A rational agent always achieves the best possible outcome because it knows the actual result of every action in advance. Hence a taxi that picks a route later blocked by an unforeseeable landslide was not rational.
S4	PEAS description of an automated taxi – Performance measure: steering, accelerator, brake, horn. Environment: roads, other traffic, pedestrians, weather. Actuators: safe, fast, legal and comfortable trip. Sensors: cameras, GPS, speedometer.
S5	Chess played without a clock is a partially observable, stochastic, episodic, static and discrete task environment.
S6	A model-based reflex agent keeps an internal state, updated using knowledge of how the world evolves and how its own actions affect it. It can therefore act sensibly even when the current percept does not reveal the full state of the world.

(a) Identify which statements are correct and which are incorrect. For each incorrect statement, quote the erroneous phrase or claim (one statement contains more than one error).

Statement	Correct / Incorrect	Erroneous phrase / claim
S1		
S2		
S3		
S4		
S5		
S6		

(b) Root cause analysis: for every error, explain the underlying misconception, i.e. which concept has been misunderstood and why the statement goes wrong.

Error (statement / part)	Root cause (misconception)

(c) Rewrite each faulty statement correctly and justify your correction. For a statement that is already correct, write “No change” and justify why.

Statement	Corrected statement	Justification
S1		
S2		
S3		
S4		
S5		
S6		

(d) Apply the corrected concepts to the following case. A self-driving taxi chooses a highway route that every available traffic report showed to be clear; an unforeseeable landslide then blocks the road.

(i) Using the four factors that decide whether an agent is rational, argue whether the taxi's choice was rational.

(ii) Name the properties of the taxi's task environment that make perfect foresight impossible.

(iii) Describe one situation in which a simple reflex taxi would behave poorly, and state what a model-based reflex taxi keeps in order to avoid it.

(i)	
(ii)	
(iii)	

(a) Identify Correct and Incorrect Statements

Statement	Correct / Incorrect	Erroneous phrase / claim
S1	Incorrect	"internal reasoning always follows the laws of logic" and "thinking rationally"
S2	Incorrect	"AI winters … occurred in the 1950s, before the 1956 Dartmouth workshop" and "workshop then revived the field"
S3	Incorrect	"always achieves the best possible outcome," "knows the actual result of every action in advance," and "was not rational" because of the unforeseeable landslide
S4	Incorrect	Performance measure and actuators are interchanged.
S5	Incorrect	"partially observable, stochastic, episodic"
S6	Correct	No erroneous phrase.

(b) Root Cause Analysis

Error	Root cause (misconception)
S1	Confuses acting humanly with thinking rationally. The Turing Test evaluates human-like conversational behaviour, not whether all internal reasoning follows formal logic.

S2 Confuses the beginning of AI with later AI winters. The Dartmouth workshop took place in 1956 and is associated with the introduction of the term “artificial intelligence.” AI winters occurred later, during periods of reduced funding and interest.

S3 Confuses rationality with omniscience. Rationality depends on available information and expected performance, not perfect knowledge of future outcomes.

S4 Confuses the purpose of PEAS components. Performance measures describe success criteria, while actuators are the mechanisms used to perform actions.

S5 Misclassifies chess. The full chessboard and all pieces are visible, moves are governed by rules, and each move affects later decisions. Chess is fully observable, deterministic, sequential and discrete.

S6 No misconception. It correctly describes the internal state and world model of a model-based reflex agent.

(c) Corrected Statements and Justifications

Statement	**Corrected statement**	**Justification**
S1	The Turing Test, proposed in 1950, evaluates whether a machine can exhibit behaviour indistinguishable from a human in conversation. It is an example of the acting humanly approach to AI.	The test focuses on observable behaviour, not whether the machine reasons using formal logic.
S2	The Dartmouth workshop took place in 1956 and is considered a foundational milestone in AI history. John McCarthy introduced the term “artificial intelligence” in connection with this workshop. AI winters occurred later, when funding and interest declined.	The AI winters did not occur before the Dartmouth workshop.
S3	A rational agent selects the action expected to maximise its performance based on its percepts, available knowledge and possible outcomes. A taxi choosing a route based on reliable traffic reports can be rational even if an unforeseeable landslide later blocks it.	Rationality does not require complete knowledge of the future.
S4	Performance measure: safe, fast, legal and comfortable trips. Environment: roads, other traffic, pedestrians and weather. Actuators: steering, accelerator, brake and horn. Sensors: cameras, GPS and speedometer.	PEAS must correctly distinguish success criteria, surroundings, action mechanisms and information-gathering devices.
S5	Chess played without a clock is a fully observable, deterministic, sequential, static and discrete task environment.	The board is visible, moves follow fixed rules, actions affect later states, no clock-

		driven changes occur, and states and actions are discrete.
S6	No change. A model-based reflex agent maintains an internal state using a model of how the world evolves and how its actions affect it.	This internal state helps the agent make decisions when a single percept does not reveal the entire world.

(d) Apply the Corrected Concepts

(i) Was the taxi's choice rational?

The taxi's choice can be considered rational if it selected the highway based on the available traffic reports and the expected performance of the journey.

The four factors determining rationality are:

1. Performance measure: Reach the destination safely, legally, comfortably and efficiently.
2. Percept sequence: All available traffic reports showed the highway was clear.
3. Knowledge of the environment: The taxi used the information and route knowledge available to it.
4. Available actions: It selected the highway route from the available route options.

The landslide was unforeseeable, so the blocked road does not automatically mean the original decision was irrational. Rationality is judged using the information available when the decision was made, not simply by the outcome.

(ii) Task-environment properties that prevent perfect foresight

- Partially observable: Sensors cannot reveal every condition of the road or every possible hazard.
- Stochastic: Uncertain events, such as sudden landslides, can affect outcomes.
- Dynamic: Traffic and road conditions can change while the taxi is travelling.

These properties make it impossible to guarantee perfect knowledge of every future event.

(iii) Simple reflex taxi versus model-based reflex taxi

Situation: A simple reflex taxi sees a clear road ahead and continues driving, even though it previously encountered a warning that the road may be closed farther along.

- Simple reflex taxi: Uses only the current percept and may fail to consider earlier information.
- Model-based reflex taxi: Maintains an internal state and a model of the environment, including relevant earlier observations, road conditions and the effects of its actions.

The model-based taxi can use that information to reconsider its route rather than blindly following a rule based only on the current percept.

Question 2: Problem-Solving Agents and Problem Formulation – Error Analysis of a Formulation

Topic: Problem-solving agents, problem formulation

A student formulates the **Missionaries and Cannibals** problem for a problem-solving agent. Three missionaries and three cannibals stand on the left bank of a river, with a boat that carries one or two people and cannot cross empty. The boat is on the left bank at the start. Missionaries must never be outnumbered by cannibals on either bank (a bank with no missionaries is safe). The objective is to move everyone to the right bank using the **fewest crossings**.

Component	Student's formulation
State representation	(M, C) = number of missionaries and cannibals on the left bank
Initial state	(3, 3)
Actions	Take the boat across with 1M, 2M, 1C, 2C or 1M + 1C
Goal test	M = C
Legality check	A state is legal if C ≤ M on the left bank (or M = 0 there)
Path cost	Number of people carried in the crossing
Note N1	The agent works in the order: search → goal formulation → problem formulation → execution.
Note N2	The state space contains 16 states, of which 10 are legal.

(a) For each component and note, state whether it is correct or incorrect; for each incorrect one, state what is wrong.

Component / Note	Correct / Incorrect	What is wrong
State representation		
Initial state		
Actions		
Goal test		
Legality check		
Path cost		

Component / Note	Correct / Incorrect	What is wrong
Note N1		
Note N2		

(b) Root cause analysis: for every error, explain the underlying cause, i.e. which principle of problem formulation or of the problem-solving agent has been violated.

Error (component / note)	Root cause

(c) Give the corrected version of every faulty component or note and justify each correction.

Component / Note	Corrected formulation	Justification

Component / Note	Corrected formulation	Justification

(d) Apply your corrected formulation.

(i) Complete the table of boat loads available from the initial state, writing each resulting state as (M, C, boat side) and whether it is legal, with the reason.

Boat load	Resulting state (M, C, boat)	Legal? Reason
1M		
2M		
1C		
2C		
1M + 1C		

(ii) Give one state that the student's legality check accepts although it is actually illegal.

(iii) Using your corrected path cost, state the cost of an optimal solution.

(ii)	
(iii)	

(a) Identify whether each component and note is correct or incorrect

Component / Note	Correct / Incorrect	What is wrong?
State representation	Incorrect	The state must include the boat's position as well as the numbers of missionaries and cannibals on the left bank.
Initial state	Incorrect	The initial state is incomplete because it does not specify the boat's position.

Actions	Correct	The listed actions cover all five possible boat loads: 1M, 2M, 1C, 2C, and 1M + 1C. They must also be restricted by passenger availability and safety.
Goal test	Incorrect	\(M=C\) does not mean everyone has crossed. The goal requires zero missionaries and zero cannibals on the left bank, with the boat on the right.
Legality check	Incorrect	It checks only the left bank. It must also check the right bank, where missionaries must not be outnumbered by cannibals.
Path cost	Incorrect	Counting people carried does not directly minimize the number of crossings. Each crossing should have a cost of 1.
Note N1	Incorrect	The correct order is goal formulation, problem formulation, search, and execution.
Note N2	Incorrect	There are 32 possible combinations when the boat side is included, before legality filtering, and 20 legal states under the standard safety constraints—not 16 states and 10 legal states.

(b) Root cause analysis

Error (component / note)	**Root cause**
State representation	Violates the complete state representation principle because the boat's position is missing.
Initial state	Violates the complete initial-state specification principle; all state variables must have starting values.
Goal test	Violates the goal specification principle because the test must describe the desired final condition, not a simple equality.
Legality check	Violates the constraint validation principle because both riverbanks must satisfy the safety rule.
Path cost	Violates the cost-function alignment principle because the objective is to minimize crossings, not the number of passengers transported.
Note N1	Violates the problem-solving agent sequence: the agent must formulate a goal and problem before searching for a solution.
Note N2	Violates the state-space enumeration principle by omitting the boat position and miscounting legal states.

(c) Corrected formulation with justification

Component / Note	**Corrected formulation**	**Justification**

State representation	(M,C,B), where M,C are the numbers on the left bank and $B \in \{L,R\}$ is the boat's side.	Represents all information needed to determine possible actions and safety.
Initial state	$(3,3,L)$	All six people and the boat begin on the left bank.
Actions	Move 1M, 2M, 1C, 2C, or 1M + 1C across the river, provided the passengers are available on the boat's current bank.	Defines valid boat loads and prevents impossible moves.
Goal test	$M=0,\ C=0,\ B=R$	Everyone must reach the right bank, and the boat must be there.
Legality check	On each bank, missionaries must be at least as numerous as cannibals whenever missionaries are present. Formally: $(M=0 \lor M\ge C)$ and $(3-M=0 \lor 3-M\ge 3-C)$.	Ensures neither bank has missionaries outnumbered by cannibals.
Path cost	Cost of each crossing = 1. Total path cost = number of crossings.	Directly optimizes the stated objective of the fewest crossings.
Note N1	Goal formulation → problem formulation → search → execution.	Establishes the correct problem-solving agent sequence.
Note N2	There are 32 possible states including boat position, of which 20 satisfy the safety constraints.	There are $4 \times 4 \times 2=32$ combinations and 10 safe (M,C) pairs for each boat side.

(d) Apply the corrected formulation

(i) Boat loads available from the initial state

Initial state: $(3,3,L)$

Boat load	**Resulting state (M,C,B)**	**Legal? Reason**
1M	$(2,3,R)$	Illegal — left bank has 2 missionaries and 3 cannibals.
2M	$(1,3,R)$	Illegal — left bank has 1 missionary and 3 cannibals.
1C	$(3,2,R)$	Legal — left bank has 3 missionaries and 2 cannibals; right bank has no missionaries.
2C	$(3,1,R)$	Legal — left bank has 3 missionaries and 1 cannibal; right bank has no missionaries.
1M + 1C	$(2,2,R)$	Legal — both banks satisfy the safety condition.

(ii) Give one state accepted by the student's legality check but actually illegal

Answer: \((M,C)=(2,1)\).

- Left bank: 2 missionaries and 1 cannibal — safe because \(1\leq2\).
- Right bank: 1 missionary and 2 cannibals — unsafe because cannibals outnumber missionaries.

Therefore, the student's legality check accepts this state even though it is actually illegal.

(iii) Cost of an optimal solution

Answer: 11 crossings.

Since the corrected path cost assigns a cost of 1 to each boat crossing, the optimal solution has a total path cost of 11.

Question 3: Searching for Solutions and Uninformed Search – Error Analysis of Search Traces

Topic: Searching for solutions, BFS, DFS, uniform-cost search

A student ran BFS, DFS and uniform-cost search (UCS) on the weighted undirected graph below. The start node is S and the goal node is G. **Conventions:** graph search with an explored set; neighbours expanded alphabetically; goal test when a node leaves the frontier; ties broken alphabetically. BFS and DFS ignore edge costs while searching, but path costs are measured with them. **BFS:** do not add a node that is already explored or already in the queue. **DFS:** use a stack with its top on the left; push children so that the alphabetically first child is popped first; do not push a node that is already explored or already on the stack. **UCS:** write entries as Node:g(parent) and replace an entry whenever a cheaper path to a frontier node is found.

Edge	Cost	Edge	Cost
S – C	3	B – G	3
S – D	3	D – E	2
A – B	2	D – G	6
A – C	6	E – G	3
B – D	4		

Student's BFS trace (queue initially [S])

Iteration	Node Dequeued	Queue after Expansion	Explored Set
1	S	[C, D]	{S}

Iteration	Node Dequeued	Queue after Expansion	Explored Set
2	C	[D, A]	{S, C}
3	D	[B, E, G, A]	{S, C, D}
4	B	[E, G, A]	{S, C, D, B}
5	E	[G, A]	{S, C, D, B, E}
6	G (goal)	[A]	{S, C, D, B, E}

BFS result: path S – D – G, cost 9, 6 nodes removed from the frontier.

Student's DFS trace (stack initially [S], top on the left)

Iteration	Node Popped	Stack after Pushing Children
1	S	[D, C]
2	D	[G, E, B, C]
3	G (goal)	[E, B, C]

DFS result: path S – D – G, cost 9, 3 nodes removed from the frontier.

Student's UCS trace

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), A:9(C), G:9(D)
5	B	7	A:9(C), G:9(D)
6	A	9	G:9(D)
7	G (goal)	9	–

UCS result: path S – D – G, cost 9, 7 nodes removed from the frontier.

Student's remarks

No.	Remark
R1	An explored set is not needed in BFS because BFS never revisits a node, so the Explored Set column is unnecessary.
R2	All three strategies returned S – D – G with cost 9, so all three are optimal and the traces are confirmed correct.
R3	DFS removed the fewest nodes from the frontier (3), so DFS is the best strategy for this graph.

(a) Examine the three traces and the remarks. List every error, giving the trace and iteration (or the remark number) and what is wrong. An error that is only carried forward counts once.

No.	Where (trace / iteration / remark)	What is wrong
1		
2		
3		
4		
5		
6		
7		

(b) Root cause analysis: for each error listed in (a), explain the rule or concept that was violated and how it led to the wrong entry or conclusion.

No.	Root cause (rule or concept violated)
1	
2	
3	

No.	Root cause (rule or concept violated)
4	
5	
6	
7	

(c) Write the corrected traces and justify each correction.

(i) BFS: the corrected queue after iteration 3 and the full order in which nodes are dequeued.

Queue after iteration 3	
Order of dequeued nodes	

(ii) DFS: the corrected trace.

Iteration	Node Popped	Stack after Pushing Children
1		
2		
3		
4		
5		

(iii) UCS: the corrected trace.

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1			
2			
3			

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
4			
5			
6			

Justification of the corrections	

(d) Apply the corrected traces.

(i) Complete the comparison table.

Strategy	Path Found	Path Cost	Nodes Removed from Frontier	Optimal Here?
BFS				
DFS				
UCS				

(ii) Use iteration 3 of the corrected BFS trace to explain why an explored set is needed in graph search.

(ii)	

(a) Identify every error in the traces and remarks

No.	Where (trace / iteration / remark)	What is wrong
1	BFS, iteration 3	Node A is already in the queue from iteration 2. It must not be added again when expanding D. The correct queue after iteration 3 is [B, E, G, A] — the student's displayed queue actually matches this, so there is no error in this entry.
2	BFS result	The BFS path is not optimal by weighted cost. BFS finds the shallowest path, which is S–D–G, with cost \(3+6=9\). A cheaper path exists: S–C–A–B–G,

		with cost \(3+6+2+3=14\); this is actually more expensive, so the claimed path cost is correct, but BFS is not guaranteed to find a minimum-cost path.
3	DFS, iteration 1	The stack after expanding S is [C, D] under the stated convention that the alphabetically first child is popped first. The student's [D, C] is incorrect.
4	DFS, iteration 2	The stack and subsequent trace are affected by the incorrect child-pushing order in iteration 1. The corrected DFS must follow the stack convention consistently.
5	UCS, iteration 3	The entry A:9(C) should be replaced by A:9(C) only if a strictly cheaper path is found; the path through D has cost \(3+4+2=9\), equal to the existing cost 9, so no replacement is needed. This entry is correct.
6	Remark R1	An explored set is still needed in BFS graph search to prevent revisiting already expanded nodes.
7	Remark R2 and R3	R2 incorrectly concludes that all strategies are optimal just because they return the same path. R3 incorrectly claims DFS is best solely because it removes fewer nodes; search quality also depends on completeness, optimality, and problem requirements.

(b) Root cause analysis

No.	Root cause (rule or concept violated)
1	BFS queue discipline: Existing frontier nodes must retain their positions. Newly discovered neighbours are appended to the end of the FIFO queue.
2	BFS completeness of trace: Every node must be accounted for in the dequeue sequence. The misplaced A changes the subsequent trace and the total number of removals.
3	DFS stack convention: The alphabetically first child must be popped first, so the stack must have C at the top after expanding S.
4	DFS trace consistency: Later expansions must follow the corrected stack order, and the path must follow the actual parent links.
5	UCS frontier update rule: When a cheaper route to a frontier node is found, its cost and parent must be updated.
6	Graph search principle: An explored set prevents repeatedly expanding states and helps avoid cycles in graph search.
7	Search evaluation principles: BFS optimizes the number of edges in an unweighted search, DFS does not guarantee optimality, and UCS finds a minimum-cost path with nonnegative edge costs. Fewer frontier removals alone do not make DFS the best strategy.

(c) Corrected traces and justifications

(i) BFS

Queue after iteration 3: [A, B, E, G]

Order of dequeued nodes: S, C, D, A, B, E, G

Iteration	Node dequeued	Queue after expansion	Explored set
1	S	[C, D]	
2	C	[D, A]	
3	D	[A, B, E, G]	
4	A	[B, E, G]	
5	B	[E, G]	
6	E	[G]	
7	G (goal)	[]	

The goal test occurs when G leaves the frontier, so G is not added to the explored set in this trace.

(ii) DFS

The stack's top is on the left. Neighbours are processed alphabetically, and already explored or stacked nodes are not pushed.

Iteration	Node popped	Stack after pushing children
1	S	[C, D]
2	C	[A, D]
3	A	[B, D]
4	B	[G, D]
5	G (goal)	[D]

Its cost is \(3+6+2+3=14\).

(iii) Uniform-cost search (UCS)

Entries are written as Node:g(parent). The frontier is ordered by increasing path cost, with alphabetical tie-breaking.

Iteration	Node expanded	\(g(n)\)	Priority queue after expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), G:8(E), A:9(C)
5	B	7	G:8(E), A:9(C)
6	G (goal)	8	—

Justification of the corrections

- BFS: A was already in the queue before D was expanded, so A stays ahead of B, E and G.
- DFS: C is popped before D, which leads to the corrected sequence S, C, A, B, G.
- UCS: When E is expanded, the route to G through E costs \(5+3=8\), which improves G's existing cost of 9. UCS replaces the old entry and parent with G:8(E).

(d) Apply the corrected traces

(i) Comparison table

Strategy	Path found	Path cost	Nodes removed from frontier	Optimal here?
BFS	S–D–G	9	7	No, not by weighted cost
DFS	S–C–A–B–G	14	5	No
UCS	S–D–E–G	8	6	Yes

(ii) Why is an explored set needed in graph search?

At iteration 3 of BFS, the algorithm expands D. Its neighbours include S and B, E and G. Since S is already explored, it must not be added to the queue again. Similarly, when A is expanded at iteration 4, C is already explored and B is already in the queue, so neither is added again.

The explored set prevents repeated expansions and unnecessary cycles, making graph search more efficient and ensuring that previously expanded nodes are not repeatedly revisited.

Question 4: Search with Partial Information – Error Analysis of Belief-State Computation

Topic: Problem formulation, search with partial information (sensorless and percept-based belief-state search)

A vacuum agent lives in a two-square world (Left square, Right square), each square being Clean (C) or Dirty (D). Actions: Left, Right and Suck (Left in the left square and Right in the right square leave the agent where it is; Suck cleans the current square). The goal is that both squares are Clean. Physical states are written (Agent, Left square, Right square):

State	Description	State	Description
S1	(L, D, D)	S5	(L, C, D)
S2	(R, D, D)	S6	(R, C, D)
S3	(L, D, C)	S7	(L, C, C)
S4	(R, D, C)	S8	(R, C, C)

Task A – sensorless agent. The agent knows it is in the Left square but nothing about the dirt, so its initial belief state is {S1, S3, S5, S7}. A student applied the plan [Suck, Right, Suck]:

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}
3	Suck	{S6, S8}

Task B – agent with a local sensor that reports [location, status of the current square]. Starting from the belief state of all eight states, the agent receives the percept [Left, Dirty], performs Suck, receives [Left, Clean], performs Right and receives [Right, Dirty]. The student's table:

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S2, S3, S6}
1	Suck	{S4, S5, S7, S8}	[Left, Clean]	{S5, S7}
2	Right	{S6, S7}	[Right, Dirty]	{S6}

Student's conclusions

No.	Conclusion
C1	Task A: the final belief state still contains two states, so the agent can never know where it is. No plan can guarantee the goal when the agent has no sensors; hence sensorless problems cannot be solved.
C2	Task B: the belief state is now the singleton {S6}, so the agent knows its location and the dirt status, and the goal has been reached.
C3	With 8 physical states there are only 8 belief states (one per physical state), so searching over belief states is no harder than searching over physical states.

(a) Examine the tables and the conclusions. List every error, giving the task and step (or the conclusion number) and what is wrong. An error that is only carried forward from an earlier wrong row counts once.

No.	Where (task / step / conclusion)	What is wrong
1		
2		
3		
4		
5		
6		
7		

(b) Root cause analysis: for each error listed in (a), explain the concept or rule of belief-state search that was misapplied.

No.	Root cause (concept or rule misapplied)
1	
2	
3	
4	
5	
6	
7	

(c) Write the corrected tables and conclusions, with justification.

(i) Task A: the corrected belief states.

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	
2	Right	
3	Suck	

(ii) Task B: the corrected belief states after each action and each percept.

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	
1	Suck		[Left, Clean]	
2	Right		[Right, Dirty]	

(iii) Corrected conclusions C1 and C2, with justification.

C1	
C2	

(d) Apply the corrected concepts.

(i) How many different belief states are possible when there are 8 physical states? Why does this make belief-state search harder than search over physical states?

(i)	

(ii) For the initial belief state {S1, S3, S5, S7}, apply the plan [Right, Suck], write the belief state after each action, and state whether it is a solution.

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Right	
2	Suck	
	Solution? Why?	

a) Identify every error in the tables and conclusions

No.	Where (task / step / conclusion)	What is wrong
1	Task A, Step 1	The result {S5, S7} is incorrect. Suck cleans the left square in S1 and S3 as well, so the correct belief state is {S5, S6, S7, S8}.
2	Task A, Step 2	The result {S6, S8} is incorrect because it is based on the incorrect belief state in Step 1.
3	Task A, Step 3	Suck on the right square cleans it. The belief state should reflect that cleaning, not remain {S6, S8}.
4	Task B, Step 0	The updated belief state {S1, S2, S3, S6} is incorrect. The percept [Left, Dirty] requires the agent to be on the left with the left square dirty, giving {S1, S3}.
5	Task B, Step 1	The predicted and updated belief states are incorrect because they are based on the wrong initial percept update.
6	Task B, Step 2	The predicted and updated belief states are incorrect because the previous belief state was wrong.
7	Conclusions C1, C2 and C3	C1 incorrectly claims sensorless problems cannot be solved; C2 incorrectly claims the goal is reached when the right square is dirty; C3 incorrectly assumes there are only 8 possible belief states.

(b) Root cause analysis

No.	Root cause (concept or rule misapplied)
1	Action image rule: Apply the action to every physical state in the belief state and take the union of all resulting states.
2	State-transition rule: Moving Right changes the agent's location but does not change the dirt status of either square.

3 Action effect rule: Suck cleans the current square; if it is already clean, the state remains unchanged.

4 Percept filtering rule: Retain only states consistent with both the reported location and the cleanliness of the current square.

5 Belief-state prediction rule: The predicted belief state must contain exactly the successors of the current belief states under the chosen action.

6 Percept update rule: After an action, filter the predicted belief state using the new percept.

7 Belief-state search principles: Sensorless search can solve some problems without knowing the exact physical state; a singleton belief state does not automatically mean the goal is reached; and belief states are subsets of physical states.

(c) Corrected tables and conclusions

(i) Task A: Corrected belief states

Step	Action	Belief state after action
0	Start	
1	Suck	
2	Right	
3	Suck	

Justification: Suck cleans the left square in Step 1. Right moves the agent to the right square in Step 2. The final Suck cleans the right square in S6 and leaves S8 unchanged. Both possibilities therefore converge to S8, where both squares are clean.

(ii) Task B: Corrected belief states

Step	Action	After action (predicted)	Percept	After percept (updated)
0	Start		[Left, Dirty]	
1	Suck		[Left, Clean]	
2	Right		[Right, Dirty]	

Justification:

- Step 0: The percept says the agent is on the left and the left square is dirty, leaving S1 and S3.
- Step 1: Suck cleans the left square. Both possible states become S5 or S7, consistent with the clean percept.
- Step 2: Right moves the agent to the right square. The dirty percept eliminates S8, leaving S6.

(iii) Corrected conclusions C1 and C2

C1 — Corrected conclusion:

Sensorless problems can sometimes be solved without sensors. The agent maintains a belief state representing all possible physical states and chooses actions that work across those possibilities. In Task A, the plan [Suck, Right, Suck] reaches the singleton belief state {S8}, guaranteeing that both squares are clean.

Justification: A sensorless agent does not need to know its exact physical state if its action sequence guarantees the goal from every state in its initial belief state.

C2 — Corrected conclusion:

The updated belief state {S6} means the agent knows it is on the right square, the left square is clean, and the right square is dirty. The goal has not been reached.

Justification: S6 represents (R, C, D), so the right square remains dirty. Knowing the exact state does not necessarily mean the goal condition is satisfied.

(d) Apply the corrected concepts

(i) How many different belief states are possible?

Answer: 256 belief states.

With 8 physical states, every physical state can either be included in or excluded from a belief state. Therefore,

This includes the empty set, although a reachable belief-state search may not use it.

Why is belief-state search harder?

- The search space grows from 8 physical states to as many as 256 belief states.
- Each action must be applied to every possible physical state in the current belief state.
- Percepts require filtering the predicted states.
- The larger search space can increase memory usage and search time.

(ii) Apply the plan [Right, Suck]

Initial belief state: {S1, S3, S5, S7}

Step	Action	Belief state after action
0	Start	
1	Right	
2	Suck	

Is it a solution? No.

After Right, the agent moves to the right square in every possible state. Suck then cleans the right square, but the left square may still be dirty.

- S4 = (R, D, C) — left square is dirty.
- S8 = (R, C, C) — both squares are clean.

Because the final belief state contains S4, the plan does not guarantee that both squares are clean.

Rubrics for Error Analysis

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Error Identification	**20**	Accurately identifies all errors with clear understanding of issues	Identifies most errors with minor omissions	Identifies limited errors; some are missed	Fails to identify key errors
Root Cause Analysis	**30**	Clearly explains underlying causes with strong technical reasoning	Explains causes with moderate clarity	Partial explanation; weak reasoning	Unable to identify causes of errors
Correction & Justification	**25**	Provides correct solutions with strong justification and logic	Provides correct corrections with some justification	Corrections are partially correct or weakly justified	Incorrect or no corrections provided
Application of Concepts	**15**	Effectively applies relevant concepts to analyze and resolve errors	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts
Clarity & Presentation	**10**	Presents analysis clearly, logically, and systematically	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria	Error Identification	Root Cause Analysis	Correction & Justification	Application of Concepts	Clarity & Presentation	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____